

Q1. DIRECTIONS *for questions 1 to 3:* Type in your answer in the input box provided below the question.

If the sum of all the odd factors of a number is $16\frac{2}{3}\%$ of the sum of all its even factors, then find the remainder when the number is divided by 8.

Q2. DIRECTIONS *for questions 1 to 3:* Type in your answer in the input box provided below the question.

If the minimum value of $\frac{x^2 + 6x + 7}{x^2 + 6x + 16}$ is L, find the value of 126L.

Q3. DIRECTIONS *for questions 1 to 3:* Type in your answer in the input box provided below the question.

Anita was walking along the edge of a wide road. As she walked along, she observed that the angle of elevation of the top of a pole, standing exactly on the other edge of the road, was 30° at two distinct points and was 45° at exactly one point. If Anita covered a distance of exactly $20\sqrt{2}$ m between the two points from where the angle was 30° , find the height of the pole (in m).

Q4. DIRECTIONS *for questions 4 to 9:* Select the correct alternative from the given choices.

Bag A contains four red and three black balls and bag B contains three red and four black balls. One ball is drawn at random from A and placed in B. Then, one ball is drawn at random from B and placed in A. What is the probability that the composition of the balls in the two bags remains unaltered?

☐ a) $\frac{29}{56}$

☐ b) $\frac{31}{56}$

☐ c) $\frac{15}{28}$

☐ d) $\frac{33}{56}$

Q5. DIRECTIONS *for questions 4 to 9:* Select the correct alternative from the given choices.
Consider the following set of equations:

$$3x + 5y + 2z = 23$$

$$5x + 6y + 7z = 44$$

$$Px + 28y + 25z = 178$$

If x , y and z are real, then which of the statements is/are not true?

- I. The set of equations will become inconsistent (have no solution) for at least one value of P .
- II. There exists a unique solution for the set of equations for every real value of P except 21.
- III. The set of equations will have more than one solution for at least one real value of P .

- ☐ a) I, II and III
- ☐ b) **Only II and III**
- ☐ c) **Only I**
- ☐ d) **None of the above**

Q6. DIRECTIONS for questions 4 to 9: Select the correct alternative from the given choices.

$$\sqrt{28 - 16\sqrt{3}} + \sqrt{28 + 16\sqrt{3}} =$$

- ☒ a) 8 ✓ Your answer is correct
- ☐ b) 16
- ☐ c) 32
- ☐ d) 4

Q7. DIRECTIONS *for questions 4 to 9:* Select the correct alternative from the given choices.

If a zero is appended to the extreme right of an eight-digit number written in the octal system (i.e., to base 8) and then the number so obtained is divided by $(25)_{10}$, then find the percentage increase/reduction in the value of the number.

- ☐ a) 50% reduction
- ☐ b) 60% reduction
- ☐ c) 68% reduction
- ☐ d) 100% increase

Q8. DIRECTIONS for questions 4 to 9: Select the correct alternative from the given choices.

If the total simple interest earned on ₹60,000, at a rate of interest of 7.5% per annum, for n years is ₹36,000, find n .

- ☐ a) 6
- ☐ b) 10
- ☐ c) 9
- ☒ d) 8 ✓ Your answer is correct

Q9. DIRECTIONS *for questions 4 to 9:* Select the correct alternative from the given choices.

Bhalu, the farmer purchased some sickles and axes for a total of ₹2910. Each sickle costed ₹150 and each axe costed ₹180. If he had purchased as many axes as the sickles he actually bought and vice versa, he would have spent less than what he actually did. Find the number of axes he purchased.

☐ a) 13

☐ b) 12

☐ c) 15

☐ d) 17

Q10. DIRECTIONS *for questions 10 and 11:* Type in your answer in the input box provided below the question.

A man of height 1.8 m walks, at a uniform speed of 14 m/min, away from a lamp post of height 6 m. The rate (in m/min) at which the length of his shadow (due to the lamp at the top of the lamp post) increases is

Q11. DIRECTIONS *for questions 10 and 11:* Type in your answer in the input box provided below the question.

A group of men can complete a work in a certain number of days. The entire group starts the work. However, exactly 16 days after the start, one-third of the group left. To complete the remaining work, the rest of the group took as many days as the entire group would have taken to complete the entire work. Find the total time (in days) that would have been taken to complete the entire work, if instead, one-fourth of the group had left after working for exactly 21 days from the start.

Q12. DIRECTIONS *for questions 12 to 15:* Select the correct alternative from the given choices.

Amar, Akbar and Antony decide to go for a cycling race, for which they first have a practice race on a racetrack OR. P and Q are points on the track between O and R such that $OP : PQ : QR$ is $1 : 2 : 3$. The ratio of the speeds with which Amar, Akbar and Antony covered the leg OP is $2 : 3 : 4$, while for the leg PQ it is $3 : 4 : 2$ and for the leg QR it is $4 : 2 : 3$. If the ratio of the speeds with which Amar covered the legs OP, PQ and QR is $2 : 3 : 4$, who among Amar, Akbar and Antony completed the practice race first?

- ☐ a) **Amar**
- ☐ b) **Akbar**
- ☐ c) **Antony**
- ☐ d) **Both Amar and Antony**

Q13. DIRECTIONS *for questions 12 to 15:* Select the correct alternative from the given choices.

If the percentage of discount offered on an article is equal in magnitude to the percentage of profit realized on selling the article, then which of the following is true regarding the Marked Price (MP) the Selling Price (SP) and the Cost Price (CP) of the article?

- ☐ a) **The SP is the arithmetic mean of the MP and the CP.**
- ☐ b) **The SP is the geometric mean of the MP and the CP.**
- ☐ c) **The SP is the harmonic mean of the MP and the CP.**
- ☐ d) **None of the above**

Q14. DIRECTIONS for questions 12 to 15: Select the correct alternative from the given choices.

Let S denote the infinite sum

$3 + 7x + 12x^2 + 18x^3 + 25x^4 + \dots$, where $|x| < 1$ and the coefficient of x^{n-1} is $\frac{1}{2}n(n+5)$, ($n = 1, 2, \dots$).

Then S equals

☐ a) $\frac{3+2x}{(1+x)^3}$

☐ b) $\frac{3-2x}{(1-x)^3}$

☐ c) $\frac{3+2x}{(1-x)^3}$

☐ d) $\frac{3-2x}{(1+x)^3}$

Q15. DIRECTIONS *for questions 12 to 15:* Select the correct alternative from the given choices.

A rice merchant has four varieties of rice – A, B, C and D – with him. He mixed the entire quantity of the rice of varieties A and B with him. With this mixture, if he mixes the entire quantity of the rice of variety C, which costs ₹20 per kg, the mixture so formed would cost ₹25 per kg. Instead, if he mixes with it the entire quantity of the rice of variety D, which costs ₹36 per kg, the mixture so formed would cost ₹31 per kg. If the entire quantity of the rice of variety D is thrice that of the rice of variety C, how much would the mixture of the entire quantity of the rice of all the four varieties cost?

- ☐ a) ₹26.5 per kg
- ☐ b) ₹28 per kg
- ☐ c) ₹28.5 per kg
- ☐ d) ₹29.5 per kg

Q16. DIRECTIONS *for question 16:* Type in your answer in the input box provided below the question.

In the front-office of a bank, there are four cash dispensing counters with a teller sitting at each of them. In how many ways can five customers stand in front of these counters, if they choose the counters at random?

Q17. DIRECTIONS *for questions 17 and 18:* Select the correct alternative from the given choices.

ABCD is a rhombus with the diagonals AC and BD intersecting at the origin on the x - y plane. If the equation of the line AB is $x - y = 1$, then the equation of the line CD is

- ☐ a) $x - y = 0$.
- ☐ b) $x + y - 1 = 0$.
- ☐ c) $x - y + 1 = 0$.
- ☐ d) None of the above

Q18. DIRECTIONS *for questions 17 and 18:* Select the correct alternative from the given choices.

If 2 is one of the roots of the quadratic equation $x^2 - (4 - \alpha)x + (\alpha + 1/\alpha) = 0$, where α is some positive number, which of the following could be the other root?

☒ a) $5/3$ ✓ **Your answer is correct**

☐ b) -1

☐ c) $1/2$

☐ d) -2

Q19. DIRECTIONS *for questions 19 and 20:* Type in your answer in the input box provided below the question.

Find the remainder when 81^{82} is divided by 17.

Q20. DIRECTIONS *for questions 19 and 20:* Type in your answer in the input box provided below the question.

Danny forgot the seven-digit telephone number of his friend, but he remembers the following information: the first three digits from the left are either 242 or 472, the digit 7 appears once and the number is an even number. If Danny were to use a trial-and-error method to reach his friend, what is the minimum number of trials that he has to make so that he can be certain of succeeding?

Q21. DIRECTIONS *for questions 21 and 22:* Select the correct alternative from the given choices.

The number of girls in a class is 60% of the number of boys in the class. In an exam, a total of 80 students of the class failed, of which 30% are girls. If 35% of the boys failed in the exam, find the percentage of girls who passed the exam.

- ☐ a) 25%
- ☐ b) 58%
- ☐ c) 42%
- ☒ d) 75% ✓ **Your answer is correct**

Q22. DIRECTIONS *for questions 21 and 22:* Select the correct alternative from the given choices.

The length, MN, and the breadth, NO, of a rectangle MNOP are 10 cm and 5 cm respectively. MN is split up into ten equal segments, by marking nine equidistant points – A, B, H and I – between M and N. Now, if ten straight lines are drawn connecting P with each of the ten points A, B, H, I and N, find the total area (in sq.cm) of all the triangles which can be observed in the resultant figure.

☐ a) 575

☐ b) 550

☐ c) 1100

☐ d) 1150